

Building a NoSQL-Powered Web Application Project

Overview:

In this course project, students will have the opportunity to design and implement a web application utilizing a NoSQL database as the primary data storage solution. Unlike traditional relational databases, NoSQL databases offer flexibility, scalability, and performance advantages, making them increasingly popular for modern web applications. Through this project, students will gain practical experience in developing a web application architecture that leverages NoSQL databases to efficiently manage and manipulate data.

Objectives:

1. Understand the fundamental concepts of NoSQL databases and their differences from traditional relational databases.
2. Explore various types of NoSQL databases such as document-based, key-value, column-family, and graph databases, and their suitable use cases.
3. Design a web application architecture that integrates a NoSQL database for data storage and retrieval.
4. Implement CRUD (Create, Read, Update, Delete) operations using NoSQL database APIs within the web application.
5. Explore data modeling techniques tailored for NoSQL databases to optimize performance and scalability.
6. Address challenges such as data consistency, concurrency control, and scalability in a NoSQL-powered web application.
7. Practice testing and debugging techniques specific to NoSQL database interactions within the web application.
8. Explore security considerations and best practices for securing a NoSQL-powered web application against common vulnerabilities.

Project Deliverables:

1. Project Proposal: A detailed proposal outlining the chosen web application idea, its target audience, and the rationale for selecting a NoSQL database.
2. System Design: Architectural diagrams illustrating the components of the web application and their interactions, with a focus on the integration of the NoSQL database.
3. Implementation: The developed web application codebase, including backend server logic, frontend user interface components, and NoSQL database integration code.
4. Documentation: Comprehensive documentation covering the project's objectives, design decisions, implementation details, and instructions for running and testing the web application.
5. Presentation: A final presentation showcasing the project, discussing design choices, implementation challenges, and lessons learned.

Project Requirements:

- Proficiency in web development technologies such as HTML, CSS, JavaScript, and a backend programming language (e.g., Node.js, Python, Ruby).
- Familiarity with basic database concepts and SQL.
- Willingness to learn and explore new technologies, particularly NoSQL databases.
- Regular attendance and participation in project meetings and feedback sessions.

By the end of this course project, students will have acquired valuable skills in web application development and database management, particularly in the context of NoSQL databases. They will be equipped to apply these skills in real-world scenarios, where the flexibility and scalability offered by NoSQL databases are increasingly in demand.

Evaluation Criteria

The project will be evaluated based on:

- Functionality: Correctness of the implemented algorithm and successful sorting of the data.
- Code quality: Readability, modularity, and proper documentation.
- Efficiency: Time complexity analysis and demonstration of efficient sorting performance.

- Presentation quality: Clarity, organization, and insightful discussion of the project outcomes.

Project Deadline

Monday 4th May 2026

Presentation will take place on Monday 4th May 2026

***Teams consists of (3) students.**